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Introduction

In this report we are using Yearly Global Land Temperature Anomalies in Degrees Celsius dataset, which covers year 1850 to 2023. After descriptive analysis of the data, we go for model specification of ARIMA models. Sets of possible ARIMA models are proposed and then are fitting using various model estimation and fitting process. The best ARIMA model(s) are then taken as result to perform forecasting in the later time.

Methodology

1. R programming software used for analysis
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Analysis

DESCRIPTIVE ANALYSIS

Reading the csv file into R

```
# Reading the data file
GlobalTemp ← read.csv("assignment2data2024.csv")
```

Figure 1

Summary statistics and plot of the dataset

```
# Descriptive Analysis
summary(GlobalTemp)
plot(GlobalTemp, type='o', ylab= 'Global Land Temperature Anomalies')
```

Figure 2

Output:

```
> summary(GlobalTemp)
      Year      Anomaly
Min.   :1850  Min.   :-0.44000
1st Qu.:1893  1st Qu.: -0.12750
Median :1936  Median :  0.00000
Mean   :1936  Mean    :  0.06218
3rd Qu.:1980  3rd Qu.:  0.23000
Max.   :2023  Max.    :  0.91000
```

Figure 3: Summary Statistics of Global Temperature dataset.

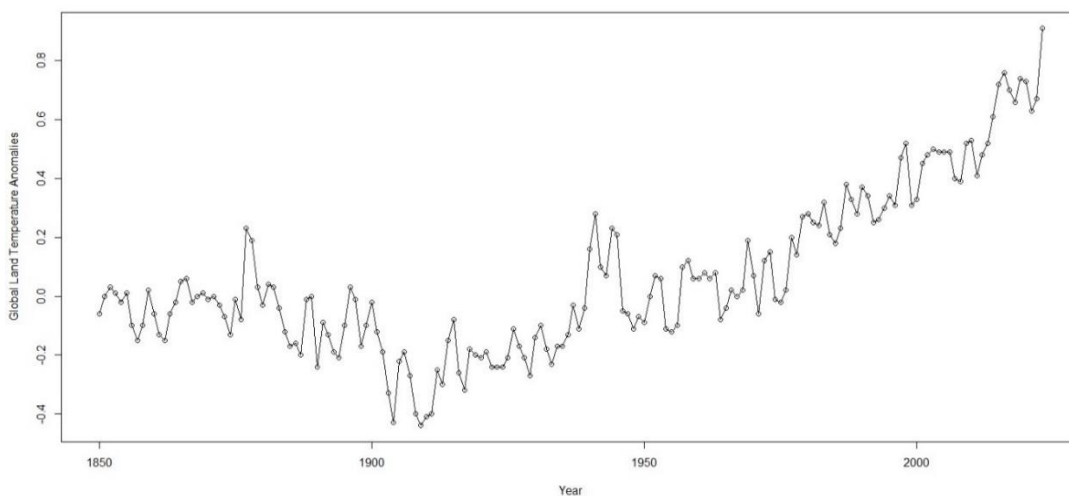


Figure 4: Plot of the dataset (Global Temperature).

Inference: Figure 3 shows that the data spans from year 1850 to 2023, the median year and mean year of this data is 1936 implying that the distribution is roughly symmetric around this value. The IQR of year has a range of 1893 to 1980, the range covers a wide range of the time-period. The anomaly variable ranges from -0.44 to 0.91, this shows fluctuations in the land temperature. The median anomaly is 0, indicating that half of the observations are positive, and half are negative anomalies. But as the mean is 0.062, so there is a slight positive deviation from the reference point. The IQR of anomaly ranges from -0.1275 to 0.23.

Converting the dataset into a time series data

```
# Convert the dataset into a time series dataset
GlobalTemp_ts <- ts(GlobalTemp$Anomaly, start = min(GlobalTemp$Year),
                    end = max(GlobalTemp$Year), frequency = 1)
```

Figure 5

Analysing raw time series data

```
# Descriptive Analysis
class(GlobalTemp_ts)
summary(GlobalTemp_ts)
plot(GlobalTemp_ts, type = 'o', ylab = 'Anomaly in Land Temperature', xlab = 'Year',
     main = "Time series plot of Global Land Temperature Anomalies in Degrees Celsius",
     col = "maroon", pch = 15)
```

Figure 6: Summary Statistics and Plot of the time series data.

Output:

```
> class(GlobalTemp_ts)
[1] "ts"
> summary(GlobalTemp_ts)
      Min.   1st Qu.   Median     Mean   3rd Qu.    Max.
-0.44000 -0.12750  0.00000  0.06218  0.23000  0.91000
```

Figure 7

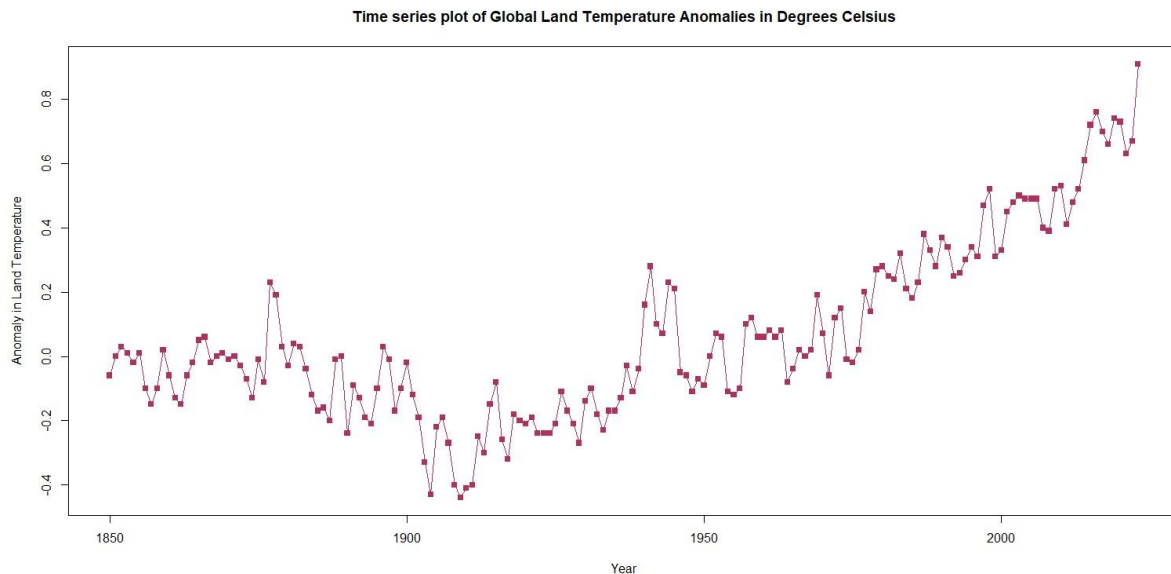


Figure 8: Time Series Plot of Global Temperature Series.

Inference: From figure 7, summary statistics of global temperature time series data shows that the median value is 0 suggesting a symmetric distribution, mean is 0.062. The interquartile range (IQR), ranges from -0.1275 to 0.230, suggesting fluctuation in temperature anomalies. The range between the first quartile -0.1275 and the third quartile 0.230 covers the middle region of the data, indicating the dispersion of temperature anomalies around the median. Severe anomalies on each side of the distribution point to times when temperatures were cold or warm as compared to the reference point.

Analysing Five Points of Interest (Figure 8)

Trend: There seems to be a quadratic upward trend in the series.

Behaviour: The series shows autoregressive i.e. AR behaviour, as in the plot we can see the successive time points.

Change point or intervention: Although there is no apparent change point in the series, but we can argue that around the year 1930 or 1940 approximately, there is some intervention happening as after that from 1950 approx. the series changes its trend and starts going upwards.

Changing Variance: By looking at figure 10 we can assume that at some time points there is some changing variance, but it is not quite evident.

Seasonality: There is no apparent evidence of seasonality in the time series.

Correlation between consecutive time points

```
# Creating first lag
orig = GlobalTemp_ts
l = zlag(GlobalTemp_ts) |
index = 2:length(orig)
cor(orig[index],l[index])

# Creating second lag
l2 = zlag(zlag(GlobalTemp_ts))
index = 3:length(l2)
cor(orig[index],l2[index])
```

Figure 9: Creating lags and then checking for correlation.

Output:

```
> cor(orig[index],l[index])
[1] 0.9399931
```

Figure 10: Value of correlation between original series and first lag.

```
> cor(orig[index],l2[index])
[1] 0.8871311
```

Figure 11: Value of correlation between the original series and second lag.

Inference: Figure 10 shows the estimated correlation coefficient, which is roughly 0.94, shows that there is autocorrelation in the time series data and that there is a strong positive correlation between the original global temperature anomaly series and its first lag. From figure 11 the correlation between original series and second lag came out to be 0.88, which is although lower than the correlation between original series and first lag, still does show strong positive linear correlation. Hence, both coefficients indicate a significant autocorrelation in the time series data.

Visual Representation of Correlation

```
# ACF plot of lags
acf(GlobalTemp_ts, lag.max = 50)

# Create a scatter plot
plot(x = l[-1], y = GlobalTemp_ts[-1], xlab = "Lag", ylab = "GlobalTemp_ts", col = "green",
     main = "Scatter Plot of Correlations between Lags (l)")
# Add second lag to the plot
points(l2[-c(1, 2)], GlobalTemp_ts[-c(1, 2)], col = "blue")
legend("topleft", legend = c("First Lag (l)", "Second Lag (l2)"), col = c("green", "blue"), pch = 1)
```

Figure 12

Output:

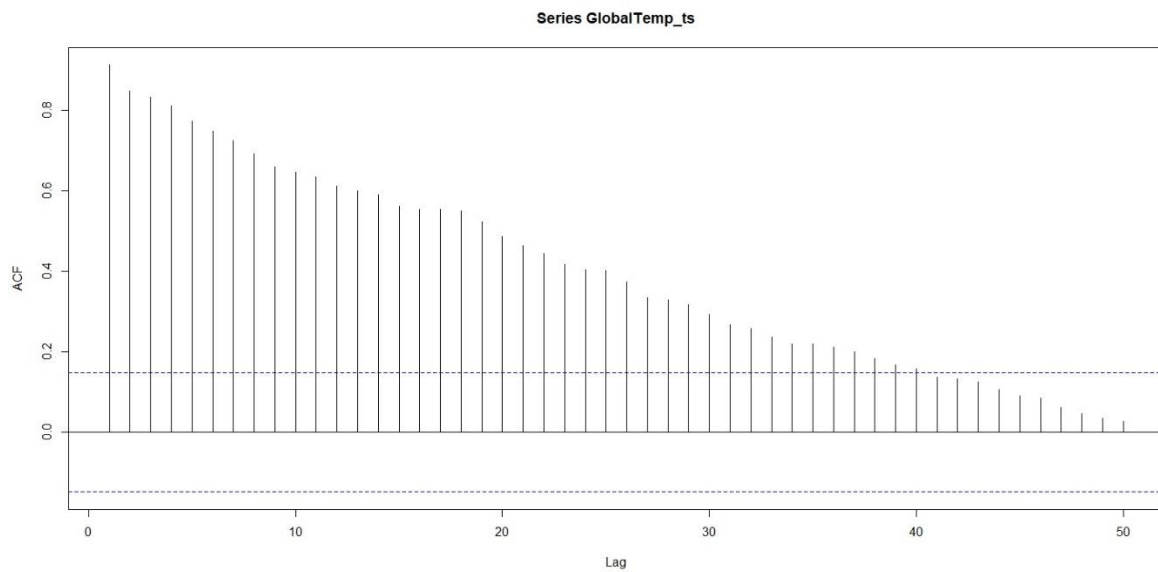


Figure 13: ACF Plot of lags.

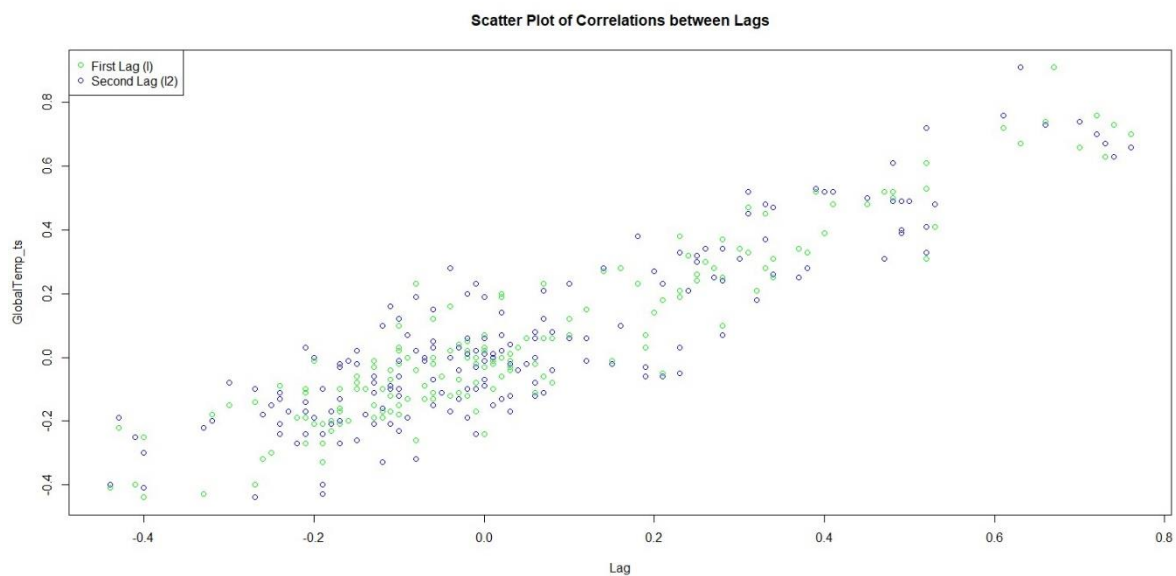


Figure 14: Scatter Plot of Correlation between lags.

Inference: Figure 13, which is of ACF plot of lags, there are significant spikes beyond the confidence interval showing presence of autocorrelation. As the lags are significantly high for almost all the data points it shows that the series is non-stationary. There are some small bumps in the decaying pattern of the ACF plot which suggests presence of seasonality in the time series data.

From figure 14, the cluster formation at certain points in the data, suggests a strong positive correlation between the two successive lags and the original data. However, the strength of positive correlation varies across data points.

MODEL SPECIFICATION

ACF and PACF of time series data

```
# ACF and PACF of the time series
par(mfrow=c(1,2))
acf(GlobalTemp_ts, main ="ACF plot of Global Land Temperature Anomaly series.")
pacf(GlobalTemp_ts, main ="PACF plot of Global Lnad Temperature Anomaly series.")
par(mfrow=c(1,1))
```

Output:

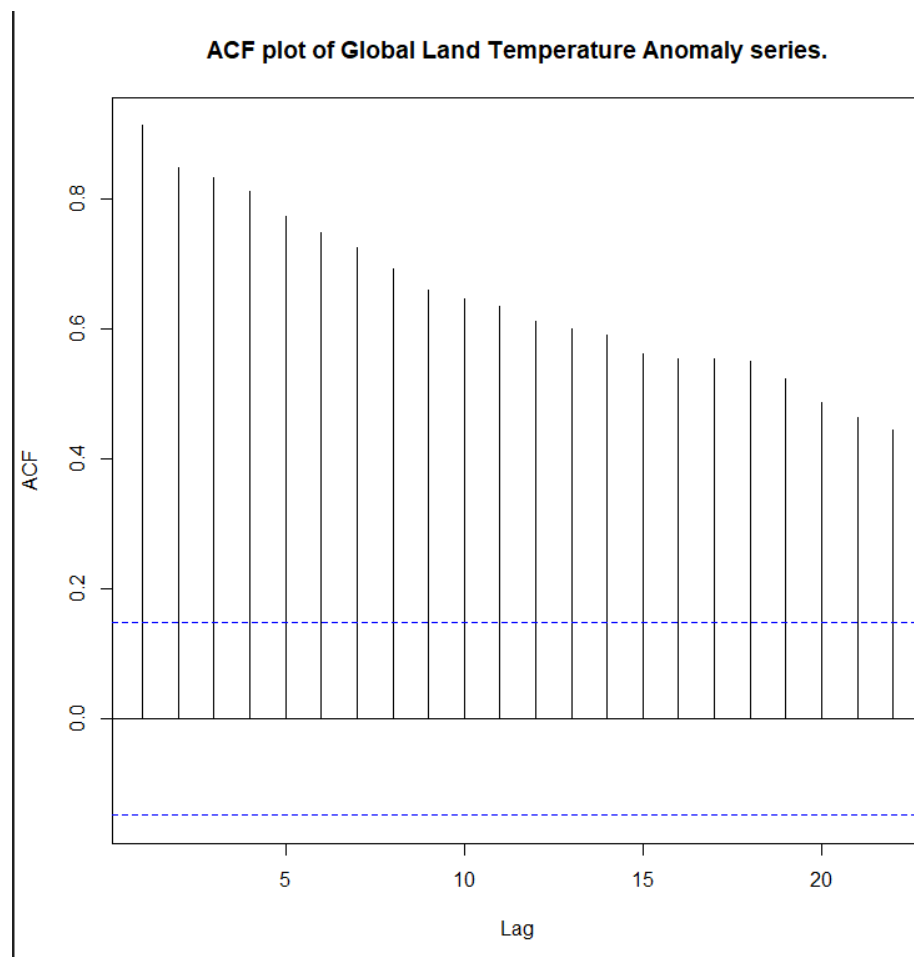


Figure 15: ACF Plot of Global Temperature time series.

PACF plot of Global Lnad Temperature Anomaly series.

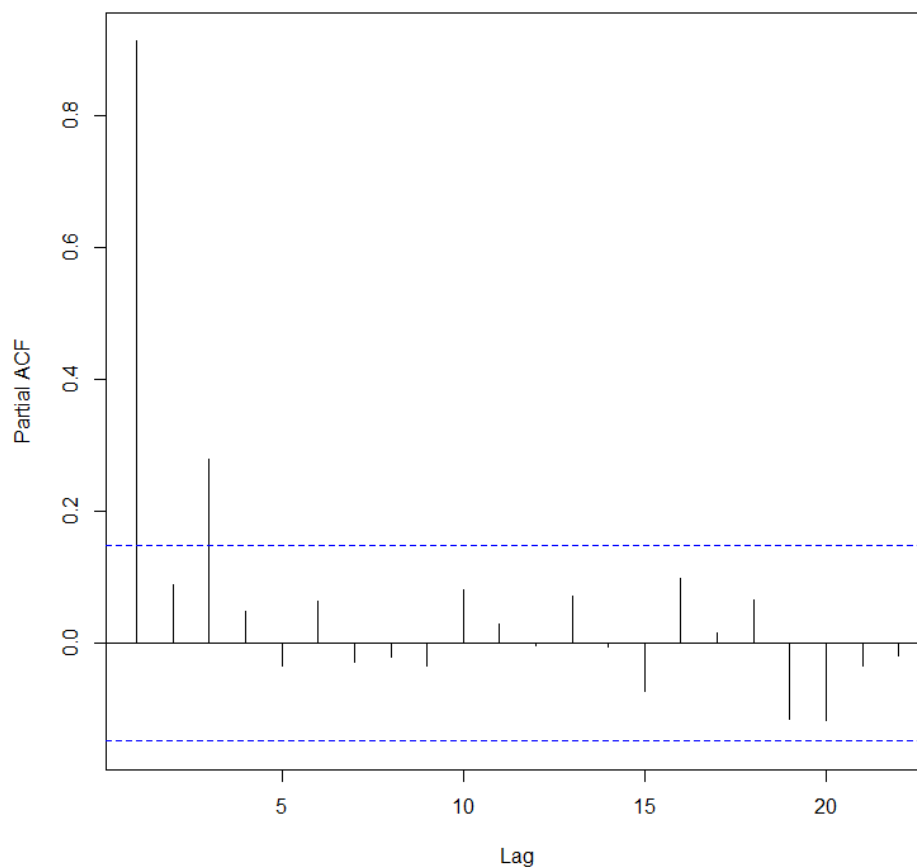


Figure 16: PACF Plot of Global Temperature time series.

Inference: The ACF plot (figure 15) has almost all significant lags with a decaying pattern which shows non-stationarity in the time series. There is a slight hint of seasonality as there is a slight wave like pattern or bumps in the slowly decaying pattern. Also, if we look at figure 16, we see one high significant lag in PACF. All these inferences suggests that the series is non-stationary.

Checking for normality

```
# QQ line
qqnorm(GlobalTemp_ts, ylab='Temperature', xlab='Normal Score'
, main= "QQ Plot of Global Temperature Time Series", col="cornsilk4")
qqline(GlobalTemp_ts, col = "darkgreen")
```

Figure 17

```
# Shapiro Wilks Test
shapiro.test(GlobalTemp_ts)
```

Figure 18

Output:

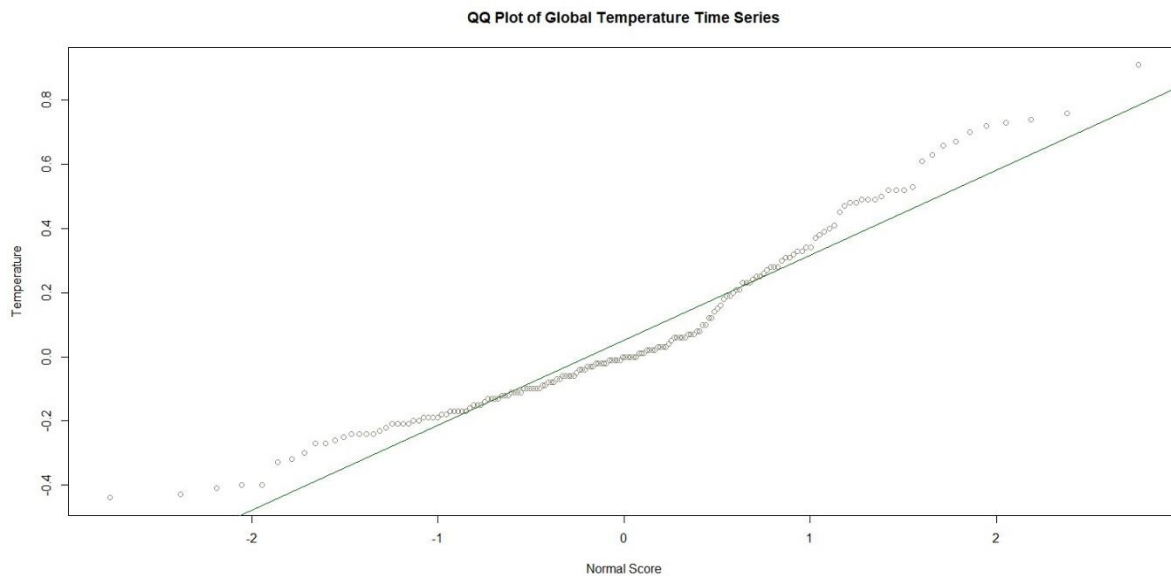


Figure 19: Q-Q Plot of Global Temperature Series.

```
Shapiro-Wilk normality test
data: GlobalTemp_ts
W = 0.94279, p-value = 1.897e-06
```

Figure 20: Shapiro-Wilk Test result.

Inference: QQ line (figure 19) doesn't fit all the point, the points are deviating from the QQ line suggesting that the time series is likely not normally distributed. From the p-value obtained from Shapiro-Wilk test (figure 20) we see that it is smaller than 0.05 hence we reject the null hypothesis and conclude that the time series do not follow a normal distribution.

Unit-root tests for stationarity

```
# Unitroot Test of the Global Temp Series
stationarity_tests(GlobalTemp_ts)
```

Figure 21: Unit-root tests.

Output:

```

> stationarity_tests(BC.GlobalTemp_ts)
Augmented Dickey-Fuller (ADF) Test:

      Augmented Dickey-Fuller Test

data:  data
Dickey-Fuller = -1.1974, Lag order = 5, p-value = 0.9044
alternative hypothesis: stationary

Kwiatkowski-Phillips-Schmidt-Shin (KPSS) Test:

      KPSS Test for Level Stationarity

data:  data
KPSS Level = 2.3416, Truncation lag parameter = 4, p-value = 0.01

Phillips-Perron (PP) Test:

      Phillips-Perron Unit Root Test

data:  data
Dickey-Fuller Z(alpha) = -12.042, Truncation lag parameter = 4, p-value = 0.4258
alternative hypothesis: stationary

```

Figure 22: Unit-Root test result.

Inference: According to ADF test p-value came out to be 0.9044 which is greater than significance level of 0.05 hence we fail to reject the null hypothesis stating that the series is most likely to be non-stationary. Figure 22 which shows the resulting p-value from KPSS test, since the p-value is smaller than the printed p-value suggesting that the p-value is even smaller than 0.01 hence, we reject the null hypothesis and state that the series is not level stationary. PP test gives out a p-value of 0.4258 which is greater than 0.05, hence we fail to reject the null hypothesis and say that the series is likely non-stationary. Since, all the three tests are consistent with each other therefore, we conclude that the series is non-stationary, and we will further have to perform transformation and differencing in the time series.

Transformation

```

# Box-Cox Transformation on the time series to minimize changing variance
GlobalTempTS ← GlobalTemp_ts + abs(min(GlobalTemp_ts)) + 0.01
BC = BoxCox.ar(GlobalTempTS)
BC$ci
lambda ← BC$lambda[which(max(BC$loglike) == BC$loglike)]
lambda
BC.GlobalTemp_ts = (GlobalTempTS^lambda-1)/lambda

```

Figure 23

Since in our time series data we have negative values, and Box-Cox transformation cannot be applied to a series with negative values, in such a case we can either use log transformation or we can add a small positive constant to the minimum value observation and add that to our original series so that all the values become positive. We perform this step before power transformation so that we don't come up with an error message for our code.

Output:

```
> BC$ci
[1] 0.9 1.1
> lambda ← BC$lambda[which(max(BC$loglike) == BC$loglike)]
> lambda
[1] 1
```

Figure 24: Lambda value and confidence interval.

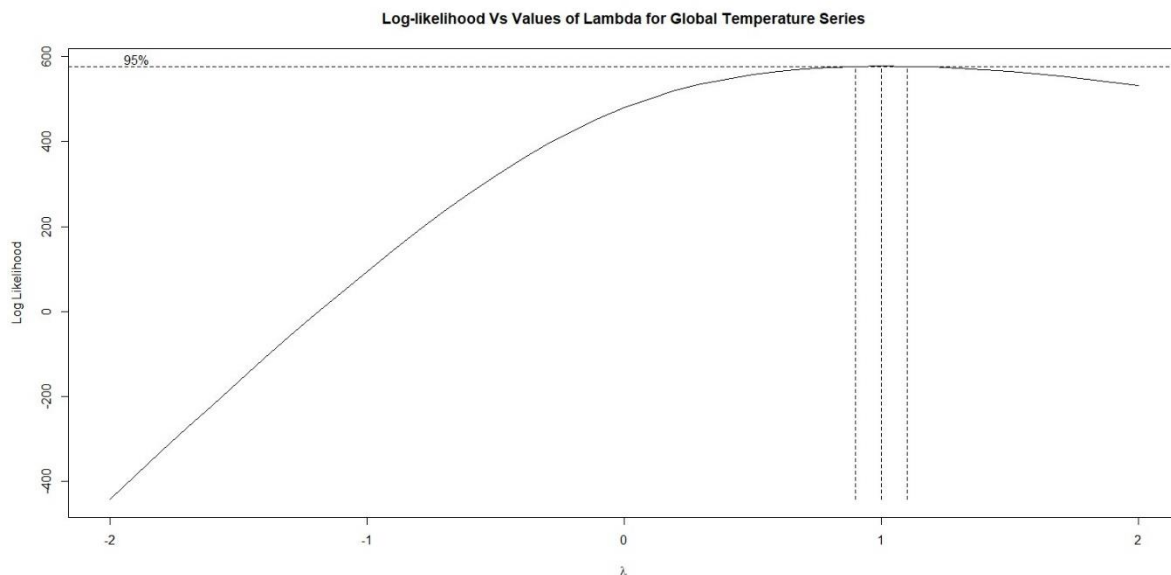


Figure 25: Box-Cox Optimum Value.

Inference: From figure 24, 95% confidence interval for lambda we will have the value of lambda between 0.9 to 1.1 and the centre of the interval corresponds to 1, which follow along the value obtained of lambda. This value will be used by Box-Cox to perform the transformation. However, when lambda is 1 so according to the formula it just simplifies to $y-1$ i.e. the transformation will just subtract 1 from each value and will lead to no significant change in the series.

Visualising Transformed series Vs Original Global Temperature Series

```
# Plotting the transformed series vs the original series
par(mfrow=c(2,1))
plot(BC.GlobalTemp_ts, type = 'o', ylab = 'Anomaly in Land Temperature', xlab = 'Year',
     main = "Transformed Time series plot of Global Land Temperature Anomalies in Degrees Celsius",
     col = "Blue", pch = 10)
plot(GlobalTemp_ts, type = 'o', ylab = 'Anomaly in Land Temperature', xlab = 'Year',
     main = "Time series plot of Global Land Temperature Anomalies in Degrees Celsius",
     col = "maroon", pch = 15)
```

Figure 26

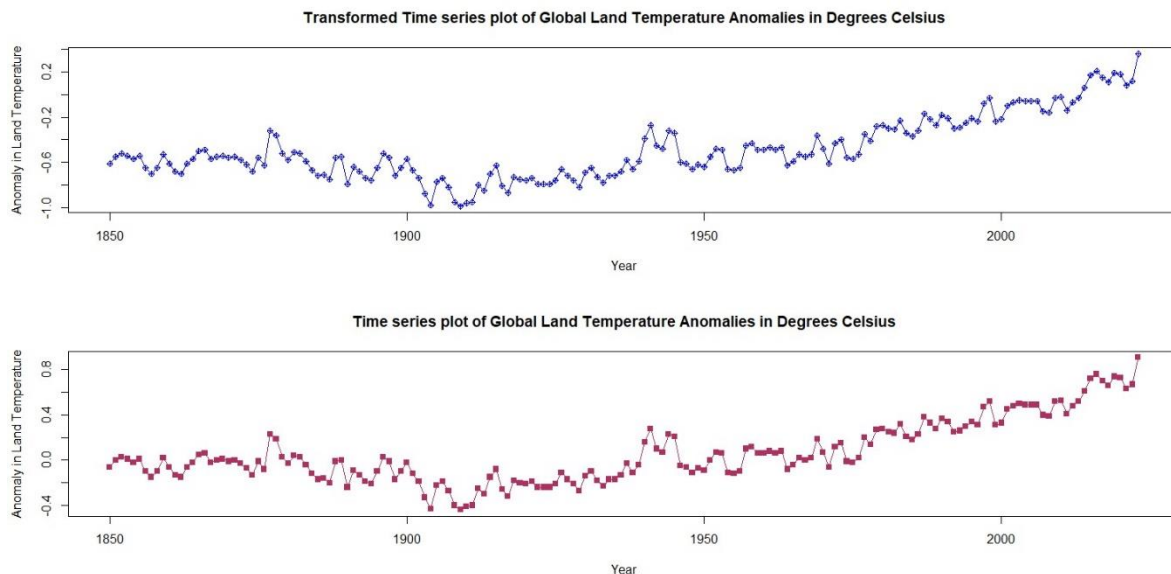


Figure 27: Time Series Plot of Original Series VS Time Series Plot of Transformed Series.

Inference: Figure 27 shows the comparison between the original time series and the transformed one, it is quite evident from the plots that transformation applied didn't make much of a difference to the time series, as discussed in the above section as well, lambda value being 1 implies that there will be no significant difference in the time series after performing transformation.

Checking for normality in the transformed time series

```
# QQ line
qqnorm(BC.GlobalTemp_ts, ylab="Box-Cox GlobalTemp_ts", xlab="Normal Scores",
      col = 'darkcyan',
      main="QQ Plot of Transformed Global Temperature Time Series")
qqline(BC.GlobalTemp_ts, col = 'darkred')
```

Figure 28

```
# Shapiro Wilk Test
shapiro.test(BC.GlobalTemp_ts)
```

Figure 29

Output:

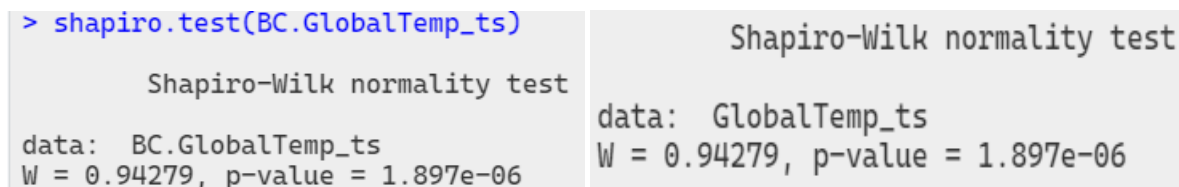


Figure 30: Shapiro-Wilk Test result comparison with raw series result.

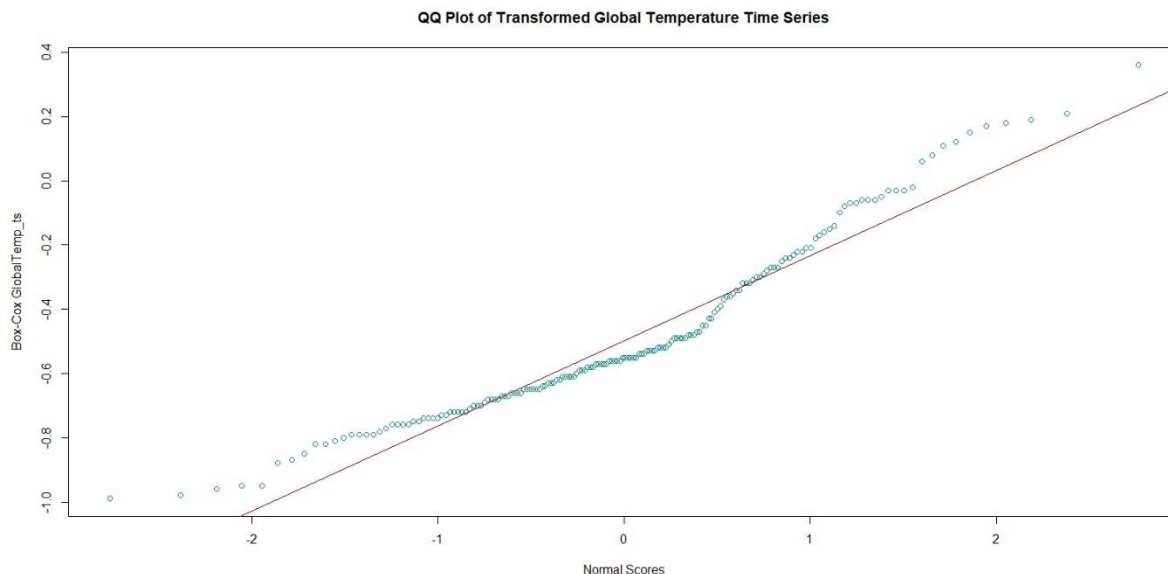


Figure 31: QQ plot of transformed time series.

Inference: Figure 30 shows the Shapiro-Wilk test result, and we make a comparison with the raw series Shapiro-Wilk test result, as evident from the resulting p-value of transformed time series we see no change in the normality after performing differencing. Now if we look at figure 31, QQ line does not fit all the data points, we can see deviation of points from the QQ line. Hence, we can state that after transformation no significant change was there in the time series data. So, for performing differencing we will be using our raw time series.

Differencing

```
# Doing first Difference
diff.GlobalTemp_ts = diff(GlobalTemp_ts)
```

Figure 32

```
# Plotting the differenced series
plot(diff.GlobalTemp_ts, type='o', pch=10, ylab='Anomaly in Land Temperature', col="coral3",
      xlab='Year', main="Time Series plot of differenced Series")
```

Figure 33

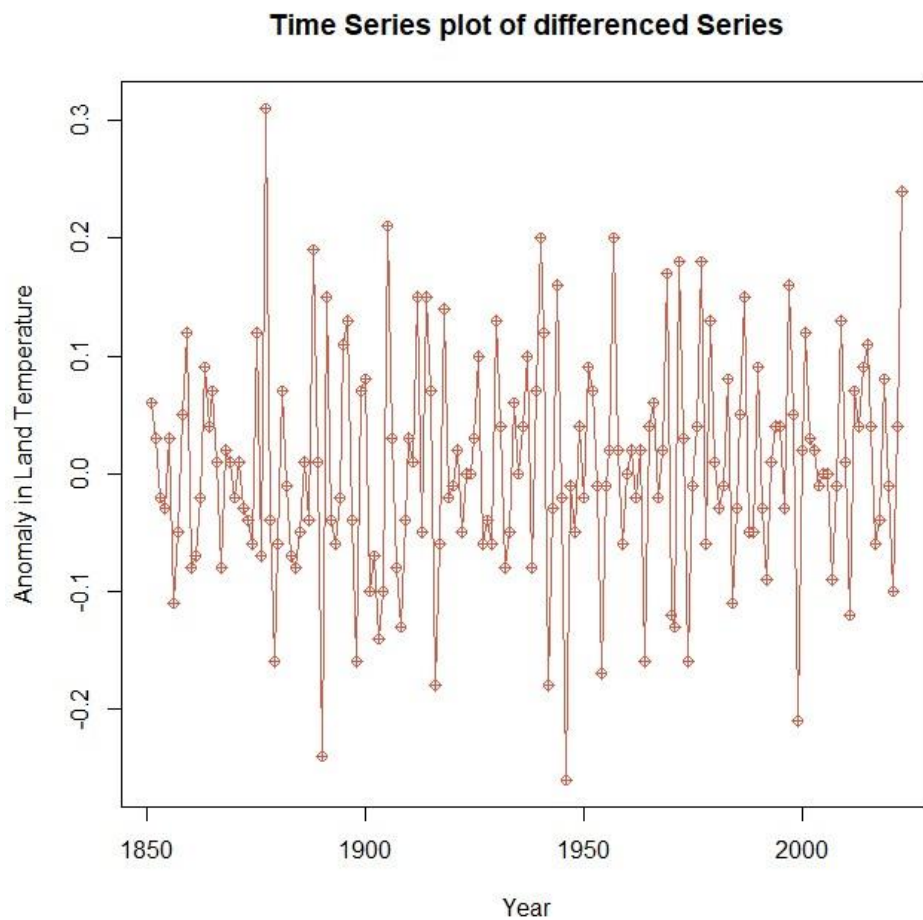


Figure 34: Differenced Time Series Plot.

Inference: The time series plot of differenced series shows no evident trend, and the mean level appears to be constant. Therefore, we can conclude that after first differencing the series is most likely to have become stationary. Although we want to support this statement with the help of stationarity tests.

Checking for stationarity after differencing

```
# Unit-root test of Differenced series
stationarity_tests(diff.GlobalTemp_ts)
```

Figure 35

Output:

```
> stationarity_tests(diff.GlobalTemp_ts)
Augmented Dickey-Fuller (ADF) Test:

      Augmented Dickey-Fuller Test

data: data
Dickey-Fuller = -7.0878, Lag order = 5, p-value = 0.01
alternative hypothesis: stationary

Kwiatkowski-Phillips-Schmidt-Shin (KPSS) Test:

      KPSS Test for Level Stationarity

data: data
KPSS Level = 0.24108, Truncation lag parameter = 4, p-value = 0.1

Phillips-Perron (PP) Test:

      Phillips-Perron Unit Root Test

data: data
Dickey-Fuller Z(alpha) = -136.61, Truncation lag parameter = 4, p-value = 0.01
alternative hypothesis: stationary
```

Figure 36: Unit-root test results.

Inference: ADF test gives a p-value of 0.01 which is less than the significance level of 0.05, hence we reject the null hypothesis and conclude that the time series is likely to be stationary. According to KPSS test with a resulting p-value of 0.1 which is greater than 0.05 we fail to reject the null hypothesis and conclude that the series is stationary. With PP test we get a p-value of 0.01, p-value being less than 0.05 we reject the null hypothesis and therefore, conclude that the time series is likely stationary. Since, all the stationarity tests are consistent with each other. Therefore, we can conclude that after performing first differencing, the time series has now become stationary. We will not be performing any further differencing as that might lead to over differencing.

After differencing, we get value of $d=1$.

Possible Models from ACF and PACF

```
# ACF and PACF plot
par(mfrow=c(2,1))
acf(diff.GlobalTemp_ts, main = "ACF plot of the first differenced Anomaly in Land Temperature series.")
pacf(diff.GlobalTemp_ts, main = "PACF plot of the first differenced Anomaly in Land Temperature series.")
```

Figure 37

Output:

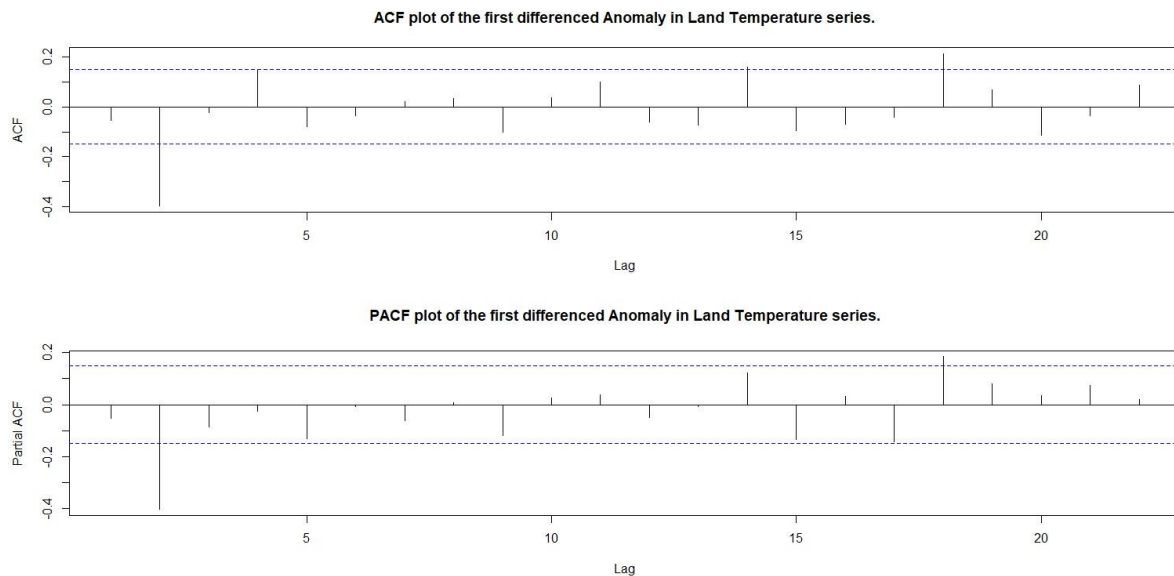


Figure 38: ACF and PACF Plot of differenced series.

Inference: From figure 38 we can see that there is 1 significant lag in PACF and 1 significant lag in ACF, although there is one near miss lag in ACF. So, we will take values of $p=1$ and $q=1, 2$.

Set of possible ARIMA models from ACF and PACF

ARIMA (1,1,1), ARIMA (1,1,2)

Possible Models from EACF

```
# EACF Plot
eacf(diff.GlobalTemp_ts)
```

Figure 39

AR/MA	0	1	2	3	4	5	6	7	8	9	10	11	12	13
0	o	x	o	o	o	o	o	o	o	o	o	o	o	o
1	o	x	o	o	o	o	o	o	o	o	o	o	o	o
2	x	o	x	o	o	o	o	o	o	o	o	o	o	o
3	x	o	x	o	o	o	o	o	o	o	o	o	o	o
4	x	x	x	o	o	o	o	o	o	o	o	o	o	o
5	o	x	o	o	o	o	x	o	o	o	o	o	o	o
6	x	x	o	x	o	o	x	o	o	o	o	o	o	o
7	x	x	o	x	x	o	o	o	o	o	o	o	o	o

Figure 40: EACF Plot

Inference: In figure 40 the first 0 which is not interrupted by x is at 0, 2. So, we take values of $p=0, 1$ and $q=2, 3$.

Set of possible ARIMA models from EACF

ARIMA (0,1,2), ARIMA (0,1,3), ARIMA (1,1,2), ARIMA (1,1,3)

Possible Models from BIC Table

```
# BIC table
res = armasubsets(y=diff.GlobalTemp_ts, nar=6, nma=6, y.name='p', ar.method='ols')
plot(res)
```

Figure 41

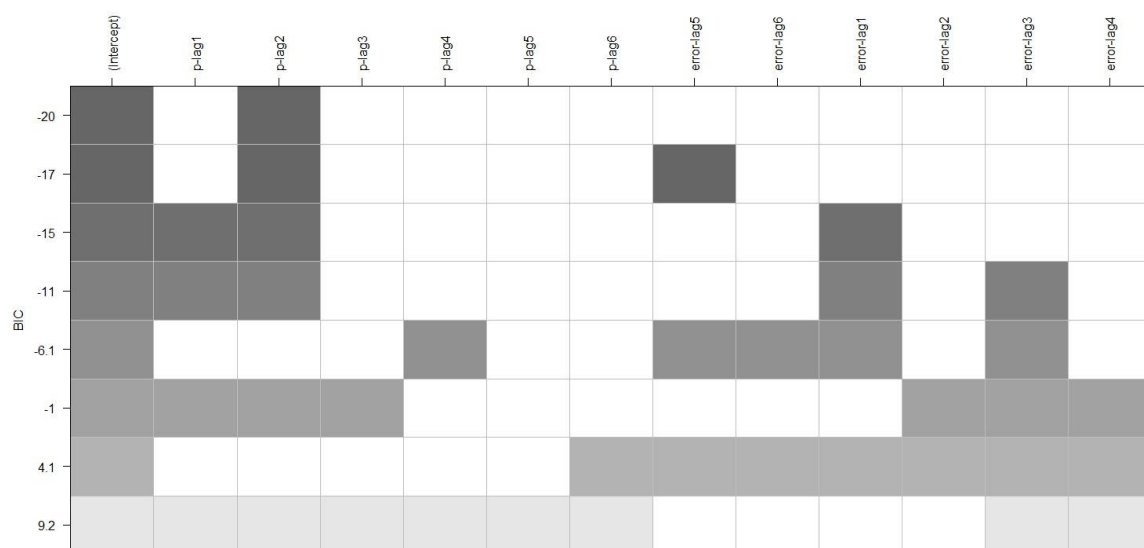


Figure 42: BIC Table

Inference: From BIC table we will take value of $p=2$ as we can see that this value is supported by first, second, third and fourth best models. Whereas for values of q , there are no parameters that are supported by the first best model but $q=5$ is at the second-best model although it is not supported by any other models, but we will take this q value into account. As per $q=1$ it is there in the third best model but if we compare the value of first best model being -20 and then for third being -15 there isn't a large gap between the two so we will also consider $q=1$.

Set of possible models from BIC Table

ARIMA (2,1,5), ARIMA (2,1,1), ARIMA (2,1,0)

All possible ARIMA models from ACF, PACF, EACF and BIC Table:

ARIMA (1,1,1), ARIMA (0,1,2), ARIMA (0,1,3), ARIMA (1,1,2),
ARIMA (1,1,3), ARIMA (2,1,5), ARIMA (2,1,1), ARIMA (2,1,0)

MODEL FITTING

Coefficient Estimation using CSS, ML and CSS-ML method.

```
# Coefficient Estimation
orders <- list(c(1, 1, 1), c(0, 1, 2), c(0, 1, 3), c(1, 1, 2), c(1, 1, 3), c(2, 1, 5), c(2, 1, 1), c(2, 1, 0))
methods <- c("CSS", "ML", "CSS-ML")
model_results <- fit_arima_models(GlobalTemp_ts, orders, methods)
model_results
```

Figure 43

Inference:

Table 1: Coefficient Estimation Summary

MODEL	CSS	ML	CSS-ML
ARIMA (1,1,1)	All coefficient significant.	All coefficient significant.	All coefficient significant.
ARIMA (0,1,2)	All coefficient significant.	All coefficient significant.	All coefficient significant.
ARIMA (0,1,3)	ma 2 significant.	ma 2 significant.	ma 2 significant.
ARIMA (1,1,2)	ma 2 significant.	ma 2 significant.	ma 2 significant.
ARIMA (1,1,3)	All coefficient significant.	All coefficient significant.	All coefficient significant.
ARIMA (2,1,5)	No coefficient significant.	No coefficient significant.	No coefficient significant.
ARIMA (2,1,1)	ar 2 significant.	ar 2 significant.	ar 2 significant.
ARIMA (2,1,0)	ar 2 significant.	ar 2 significant.	ar 2 significant.

From table 1 we find that ARIMA (0,1,2), ARIMA (1,1,1), ARIMA (1,1,3) have all coefficients significant, and the result is consistent with CSS, ML and CSS-ML model estimation method.

AIC and BIC score of ARIMA Models

```
sort.score(AIC(model_111_ml, model_012_ml, model_013_ml, model_112_ml, model_113_ml,
               model_215_ml, model_211_ml, model_210_ml), score = "aic")
sort.score(BIC(model_111_ml, model_012_ml, model_013_ml, model_112_ml, model_113_ml,
               model_215_ml, model_211_ml, model_210_ml), score = "bic")
```

Figure 44

Output:

```
> sort.score(AIC(model_111_ml,model_012_ml,model_013_ml,model_112_ml,model_113_ml,
+               model_215_ml,model_211_ml,model_210_ml), score = "aic")
      df      AIC
model_113_ml  5 -354.6224
model_211_ml  3 -353.2793
model_210_ml  3 -353.2793
model_012_ml  3 -352.3256
model_013_ml  4 -350.7104
model_112_ml  4 -350.5250
model_215_ml  8 -348.1019
model_111_ml  3 -338.5562
```

Figure 45: AIC Score of all possible ARIMA models.

```
> sort.score(BIC(model_111_ml,model_012_ml,model_013_ml,model_112_ml,model_113_ml,
+               model_215_ml,model_211_ml,model_210_ml), score = "bic" )
      df      BIC
model_211_ml  3 -343.8194
model_210_ml  3 -343.8194
model_012_ml  3 -342.8657
model_113_ml  5 -338.8560
model_013_ml  4 -338.0972
model_112_ml  4 -337.9119
model_111_ml  3 -329.0963
model_215_ml  8 -322.8756
```

Figure 46: BIC Score of all possible ARIMA models.

Inference: Figure 45 shows us the AIC score of all possible models, we can see that the least score is of ARIMA (1,1,3) although if we look at ARIMA (2,1,1) and ARIMA (2,1,0) they have the same AIC score, and the score is just having one point difference from ARIMA (1,1,3). So, from AIC score we will consider:

ARIMA (1,1,3)

ARIMA (2,1,0)

ARIMA (2,1,1)

From figure 46 we can see the BIC score of models, least score is -343.8194 of ARIMA (2,1,1) also ARIMA (2,1,0) has the same BIC score. ARIMA (0,1,2) has a BIC score of -342.8657 which approximately just 1 point more than the first two models. Hence, we will consider:

ARIMA (2,1,1)

ARIMA (2,1,0)

ARIMA (0,1,2)

AIC and BIC score are consistent for ARIMA (2,1,1) and ARIMA (2,1,0).

Error Measures of ARIMA models

```
# Error measures
Smodel_111_css <- accuracy(model_111_css)[1:7]
Smodel_012_css <- accuracy(model_012_css)[1:7]
Smodel_013_css <- accuracy(model_013_css)[1:7]
Smodel_112_css <- accuracy(model_112_css)[1:7]
Smodel_113_css <- accuracy(model_113_css)[1:7]
Smodel_215_css <- accuracy(model_215_css)[1:7]
Smodel_211_css <- accuracy(model_211_css)[1:7]
Smodel_210_css <- accuracy(model_210_css)[1:7]
df.Smodels <- data.frame(
  rbind(Smodel_111_css, Smodel_012_css, Smodel_013_css,
        Smodel_112_css, Smodel_113_css, Smodel_215_css, Smodel_211_css, Smodel_210_css)
)
colnames(df.Smodels) <- c("ME", "RMSE", "MAE", "MPE", "MAPE",
  "MASE", "ACF1")
rownames(df.Smodels) <- c("ARIMA(1,1,1)", "ARIMA(0,1,2)", "ARIMA(0,1,3)",
  "ARIMA(1,1,2)", "ARIMA(1,1,3)", "ARIMA(2,1,5)", "ARIMA(2,1,1)", "ARIMA(2,1,0)")
round(df.Smodels, digits = 3)
```

Figure 47

Output:

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1
ARIMA(1,1,1)	0.009	0.089	0.068	NaN	Inf	0.953	0.102
ARIMA(0,1,2)	0.010	0.086	0.065	NaN	Inf	0.899	0.004
ARIMA(0,1,3)	0.011	0.085	0.065	NaN	Inf	0.901	-0.023
ARIMA(1,1,2)	0.010	0.085	0.064	NaN	Inf	0.895	-0.008
ARIMA(1,1,3)	0.010	0.084	0.064	NaN	Inf	0.894	-0.022
ARIMA(2,1,5)	0.009	0.084	0.062	NaN	Inf	0.866	-0.010
ARIMA(2,1,1)	0.008	0.085	0.063	NaN	Inf	0.873	-0.001
ARIMA(2,1,0)	0.007	0.085	0.062	Inf	Inf	0.867	-0.042

Figure 48: Error Measures of all possible ARIMA models.

Inference: In figure 48 we see that mean percentage error (MPE) is showing NaN i.e. not a number and mean absolute percentage error (MAPE) is showing inf i.e. infinity these values are coming due to the presence of zero (0) in our time series data. We can do some alteration in the time series to get rid of those zeros so that we don't get NaN or inf values, but doing so might induce some bias in our analysis. We can also see that other error measures such as ME, RMSE, MAE etc. are having very close values for almost all ARIMA models. So, we will take those other values into account and ignore MPE and MAPE as even after we might address the zeros in the series there is a chance that MPE and MAPE will also be having close values for all the models and hence wouldn't add anything more to the analysis. Therefore, after considering ME, RMSE, MAE and ACF1 error measures of all our possible ARIMA models, we will consider ARIMA (2,1,0), as it is having significantly small errors as compared to other models, ARIMA (1,1,3) also has small errors, also ARIMA (2,1,5) has smallest RMSE, MAE error values. Hence, we have come up the following ARIMA models:

ARIMA (2,1,0), ARIMA (1,1,3), ARIMA (2,1,5).

Result

After performing all the analysis steps, i.e. descriptive analysis of the time series, model specification and model fitting we get the following best fit ARIMA models based on each model fitting method.

Table 2: ARIMA models after Model fitting.

Model Fitting Method used	ARIMA models
Parameter estimation	ARIMA (0,1,2), ARIMA (1,1,1), ARIMA (1,1,3)
AIC and BIC score	ARIMA (2,1,1), ARIMA (2,1,0)
Error Measures	ARIMA (2,1,0), ARIMA (1,1,3), ARIMA (2,1,5)

By looking at table 2, ARIMA (1,1,3) all the parameters are significant for this model, error measures also came out to significantly low, as per AIC and BIC score, the AIC score for ARIMA (1,1,3) was the lowest (-354.6224) however it wasn't the same with BIC score (-338.8560). But after considering parameter estimation, error measures and AIC score, we can state that ARIMA (1,1,3) is a relatively significant model. ARIMA (2,1,0) model fitting was consistent with AIC, BIC score and Error measures although in parameter estimation it had only one coefficient significant which was ar 2. However, we will consider this model as well because overall it was relatively significant and is also a smaller model.

Best fit ARIMA models:

ARIMA (1,1,3) and ARIMA (2,1,0)

We will over parameterize our best fit models so that we can explore a wider range of possible ARIMA models. In some cases, neighbouring models seems to be a better fitting ARIMA model.

Over parameterizing ARIMA (1,1,3) and ARIMA (2,1,0)

By over-parameterizing these models, we get.

ARIMA (2,1,3), ARIMA (1,1,4), ARIMA (3,1,0) and ARIMA (2,1,1)

```
orders <- list(c(2,1,3),c(1,1,4),c(3,1,0),c(2,1,1))
over_parameterised_model <- fit_arima_models(GlobalTemp_ts, orders, methods)
over_parameterised_model
```

Figure 49

Table 3: Coefficient estimation of over-parameterized models.

Model	CSS	ML	CSS-ML
ARIMA (2,1,3)	ar 1, ma 1, ma 3 significant.	ar 1, ma 1, ma 3 significant.	ar 1, ma 1, ma 3 significant.
ARIMA (1,1,4)	ma 2 significant.	ma 2 significant.	ma 2 significant.
ARIMA (3,1,0)	ar 2 significant.	ar 2 significant.	ar 2 significant.
ARIMA (2,1,1)	ar 2 significant.	ar 2 significant.	ar 2 significant.

After over-parameterizing our best fit ARIMA models, we can conclude by looking at table 3 that none of those over-parameterized models have all coefficients significant. Therefore, we won't perform any other model fitting technique to these models. Hence, we stick to our previously selected best ARIMA models, which are as follows:

ARIMA (1,1,3)

ARIMA (2,1,0)

Next step will be model validation and then further on using the best model for forecasting.

Reference

Dataset Used:

https://www.ncei.noaa.gov/access/monitoring/climate-at-a-glance/global/time-series/globe/land/ytd/12/1850-2023?trend=true&trend_base=10&begtrendyear=1850&endtrendyear=2023

Reading Material

Module 1:

https://rmit.instructure.com/courses/124176/files/36179093/download?download_frd=1

https://rmit.instructure.com/courses/124176/files/37065155/download?download_frd=1

Module 3:

https://rmit.instructure.com/courses/124176/files/36179092/download?download_frd=1

https://rmit.instructure.com/courses/124176/files/37162689/download?download_frd=1

Module 4:

https://rmit.instructure.com/courses/124176/files/36179098/download?download_frd=1

https://rmit.instructure.com/courses/124176/files/37260017/download?download_frd=1

Module 5:

https://rmit.instructure.com/courses/124176/files/36179219/download?download_frd=1

https://rmit.instructure.com/courses/124176/files/37445945/download?download_frd=1

Module 6:

https://rmit.instructure.com/courses/124176/files/36178867/download?download_frd=1

https://rmit.instructure.com/courses/124176/files/37562403/download?download_frd=1

R code reference:

https://rmit.instructure.com/courses/124176/files/37079799/download?download_frd=1

https://rmit.instructure.com/courses/124176/files/37257888/download?download_frd=1

https://rmit.instructure.com/courses/124176/files/37445488/download?download_frd=1

https://rmit.instructure.com/courses/124176/files/37562032/download?download_frd=1

https://rmit.instructure.com/courses/124176/files/37748470/download?download_frd=1

R colour names:

<https://r-graph-gallery.com/42-colors-names.html>

How to write R functions:

https://users.phhp.ufl.edu/rlp176/Courses/PHC6089/R_notes/functions.html

<https://youtu.be/p8tAQx7ijXE?si=TVwfLuOJSb2XXG9Q>

Sort Score Function:

https://rmit.instructure.com/courses/124176/files/36179115/download?download_frd=1

Appendix

Clearing R environment and loading necessary packages:

```
rm(list=ls())
install.packages("TSA")
library(TSA)
install.packages("fUnitRoots")
library(fUnitRoots)
library(tseries)
library(lmtest)
library(forecast)
library(dplyr)
```

Setting up the directory

```
# setting up the working directory
setwd("C:/Users/shamb/OneDrive/Desktop/Time Series Analysis/Assignment 2")
```

Model Estimation (CSS, ML, CSS-ML) values:

```
> model_results
$model_111_CSS

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1  0.483274   0.110230   4.3842 1.164e-05 ***
ma1 -0.772055   0.075674 -10.2024 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$model_111_ML

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1  0.50331   0.11182   4.5012 6.758e-06 ***
ma1 -0.78487   0.07365 -10.6568 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$model_111_CSS-ML

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1  0.503412   0.111791   4.5031 6.696e-06 ***
ma1 -0.784973   0.073598 -10.6657 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
$model_012_CSS
z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ma1 -0.155660   0.067130 -2.3188   0.02041 *
ma2 -0.380127   0.062131 -6.1182 9.466e-10 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$model_012_ML
z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ma1 -0.154653   0.067355 -2.2961   0.02167 *
ma2 -0.376908   0.062077 -6.0716 1.267e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$`model_012_CSS-ML`
z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ma1 -0.154685   0.067338 -2.2971   0.02161 *
ma2 -0.377000   0.062068 -6.0739 1.248e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
$model_013_CSS
z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ma1 -0.120979   0.087093 -1.3891   0.1648
ma2 -0.380308   0.061678 -6.1660 7.002e-10 ***
ma3 -0.053368   0.083661 -0.6379   0.5235
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$model_013_ML
z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ma1 -0.121311   0.086910 -1.3958   0.1628
ma2 -0.377091   0.061659 -6.1158 9.609e-10 ***
ma3 -0.051400   0.083107 -0.6185   0.5363
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$`model_013_CSS-ML`
z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ma1 -0.121301   0.086912 -1.3957   0.1628
ma2 -0.377080   0.061660 -6.1155 9.624e-10 ***
ma3 -0.051447   0.083107 -0.6191   0.5359
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

\$model_112_CSS

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)
ar1	0.063647	0.153936	0.4135	0.6793
ma1	-0.202697	0.134390	-1.5083	0.1315
ma2	-0.370220	0.067526	-5.4826	4.19e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

\$model_112_ML

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)
ar1	0.070015	0.155948	0.4490	0.6535
ma1	-0.207702	0.136249	-1.5244	0.1274
ma2	-0.366988	0.067420	-5.4433	5.231e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

\$`model\_112\_CSS-ML`

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)
ar1	0.070052	0.155949	0.4492	0.6533
ma1	-0.207737	0.136254	-1.5246	0.1274
ma2	-0.366972	0.067424	-5.4428	5.245e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

\$model_113_CSS

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)
ar1	-0.942774	0.047472	-19.860	< 2.2e-16 ***
ma1	0.847097	0.079913	10.600	< 2.2e-16 ***
ma2	-0.514501	0.081849	-6.286	3.258e-10 ***
ma3	-0.440719	0.067183	-6.560	5.382e-11 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

\$model_113_ML

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)
ar1	-0.958601	0.039213	-24.4458	< 2.2e-16 ***
ma1	0.869861	0.077785	11.1829	< 2.2e-16 ***
ma2	-0.510864	0.082772	-6.1720	6.745e-10 ***
ma3	-0.449871	0.067227	-6.6918	2.205e-11 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

\$`model\_113\_CSS-ML`

z test of coefficients:

	Estimate	Std. Error	z value	Pr(> z)
ar1	-0.958563	0.039123	-24.5010	< 2.2e-16 ***
ma1	0.870269	0.077771	11.1902	< 2.2e-16 ***
ma2	-0.510305	0.082912	-6.1548	7.517e-10 ***
ma3	-0.449688	0.067245	-6.6873	2.274e-11 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
$model_215_CSS
```

```
z test of coefficients:
```

	Estimate	Std. Error	z value	Pr(> z)
ar1	-0.169864	0.913885	-0.1859	0.8525
ar2	-0.078792	0.350468	-0.2248	0.8221
ma1	0.051752	0.912789	0.0567	0.9548
ma2	-0.382098	0.308573	-1.2383	0.2156
ma3	-0.115723	0.441337	-0.2622	0.7932
ma4	0.129309	0.195891	0.6601	0.5092
ma5	-0.074991	0.159942	-0.4689	0.6392

```
$model_215_ML
```

```
z test of coefficients:
```

	Estimate	Std. Error	z value	Pr(> z)
ar1	-0.004272	1.141425	-0.0037	0.9970
ar2	0.015289	0.540123	0.0283	0.9774
ma1	-0.113076	1.139344	-0.0992	0.9209
ma2	-0.452397	0.442003	-1.0235	0.3061
ma3	-0.036358	0.548360	-0.0663	0.9471
ma4	0.172786	0.286748	0.6026	0.5468
ma5	-0.092927	0.164863	-0.5637	0.5730

```
$`model_215_CSS-ML`
```

```
z test of coefficients:
```

	Estimate	Std. Error	z value	Pr(> z)
ar1	-0.031383	1.202914	-0.0261	0.9792
ar2	0.005499	0.550264	0.0100	0.9920
ma1	-0.085985	1.201219	-0.0716	0.9429
ma2	-0.445744	0.444318	-1.0032	0.3158
ma3	-0.049316	0.578739	-0.0852	0.9321
ma4	0.167060	0.295233	0.5659	0.5715
ma5	-0.089518	0.175070	-0.5113	0.6091

```
$model_211_CSS

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1  0.147460   0.192159  0.7674   0.4429
ar2 -0.399352   0.077451 -5.1562 2.52e-07 ***
ma1 -0.273855   0.214609 -1.2761   0.2019
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$model_211_ML

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1  0.148792   0.194411  0.7653   0.4441
ar2 -0.395596   0.077386 -5.1120 3.188e-07 ***
ma1 -0.273818   0.217564 -1.2586   0.2082
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$model_211_CSS-ML`

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1  0.148180   0.194455  0.7620   0.4460
ar2 -0.395649   0.077342 -5.1156 3.128e-07 ***
ma1 -0.273026   0.217648 -1.2544   0.2097
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
$model_210_CSS

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1 -0.078685   0.070804 -1.1113   0.2664
ar2 -0.414783   0.070755 -5.8622 4.567e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$model_210_ML

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1 -0.077561   0.070832 -1.0950   0.2735
ar2 -0.411121   0.070413 -5.8387 5.261e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$model_210_CSS-ML`

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1 -0.077560   0.070832 -1.0950   0.2735
ar2 -0.411120   0.070413 -5.8387 5.261e-09 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Model Name defining for further use in AIC, BIC and error measures:

```
# Model name defining for AIC, BIC and error measures

model_111_css = Arima(GlobalTemp_ts,order=c(1,1,1),method='CSS')
model_111_ml = Arima(GlobalTemp_ts,order=c(1,1,1),method='ML')

model_013_css = Arima(GlobalTemp_ts,order=c(0,1,3),method='CSS')
model_013_ml = Arima(GlobalTemp_ts,order=c(0,1,3),method='ML')

model_012_css = Arima(GlobalTemp_ts,order=c(0,1,2),method='CSS')
model_012_ml = Arima(GlobalTemp_ts,order=c(0,1,2),method='ML')

model_112_css = Arima(GlobalTemp_ts,order=c(1,1,2),method='CSS')
model_112_ml = Arima(GlobalTemp_ts,order=c(1,1,2),method='ML')

model_113_css = Arima(GlobalTemp_ts,order=c(1,1,3),method='CSS')
model_113_ml = Arima(GlobalTemp_ts,order=c(1,1,3),method='ML')

model_215_css = Arima(GlobalTemp_ts,order=c(2,1,5),method='CSS')
model_215_ml = Arima(GlobalTemp_ts,order=c(2,1,5),method='ML')

model_211_css = Arima(GlobalTemp_ts,order=c(2,1,1),method='CSS')
model_211_ml = Arima(GlobalTemp_ts,order=c(2,1,1),method='ML')

model_210_css = Arima(GlobalTemp_ts,order=c(2,1,0),method='CSS')
model_210_ml = Arima(GlobalTemp_ts,order=c(2,1,0),method='ML')
```

AIC and BIC sorting function:

```
# Function for sorting AIC and BIC output
sort.score <- function(x, score = c("bic", "aic")){
  if (score == "aic"){
    x[with(x, order(AIC)),]
  } else if (score == "bic") {
    x[with(x, order(BIC)),]
  } else {
    warning('score = "x" only accepts valid arguments ("aic","bic")')
  }
}
```

Function for coefficient estimation:

```
# Function for coefficient estimation
fit_arma_models <- function(data, orders, methods) {
  results <- list()

  for (order in orders) {
    for (method in methods) {
      model <- Arima(data, order = order, method = method)
      test_result <- lmtest::coefestest(model)
      result_name <- paste("model_", paste(order, collapse = ""), "_", method, sep = "")
      results[[result_name]] <- test_result
    }
  }

  return(results)
}
```

Function for stationarity tests:

```
# Stationarity test function
stationarity_tests <- function(data) {

  adf_result <- adf.test(data)
  cat("Augmented Dickey-Fuller (ADF) Test:\n")
  print(adf_result)

  kpss_result <- kpss.test(data)
  cat("\nKwiatkowski-Phillips-Schmidt-Shin (KPSS) Test:\n")
  print(kpss_result)

  pp_result <- pp.test(data)
  cat("\nPhillips-Perron (PP) Test:\n")
  print(pp_result)
}
```

Over-parameterized models coefficient test:

```
> over_parameterised_model
$model_213_CSS

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1 -0.71115    0.27835  -2.5548 0.010624 *
ar2 -0.26928    0.15937  -1.6897 0.091094 .
ma1  0.59738    0.26847   2.2252 0.026070 *
ma2 -0.25097    0.14362  -1.7474 0.080564 .
ma3 -0.38770    0.12859  -3.0151 0.002569 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$model_213_ML

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1 -0.76461    0.25791  -2.9646 0.0030310 **
ar2 -0.28027    0.17727  -1.5810 0.1138770
ma1  0.64837    0.24063   2.6945 0.0070495 **
ma2 -0.24217    0.15740  -1.5385 0.1239177
ma3 -0.40503    0.11308  -3.5817 0.0003413 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$model_213_CSS-ML`

z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1 -0.76412    0.25815  -2.9599 0.0030771 **
ar2 -0.28008    0.17716  -1.5809 0.1138919
ma1  0.64794    0.24091   2.6895 0.0071561 **
ma2 -0.24227    0.15730  -1.5402 0.1235162
ma3 -0.40488    0.11319  -3.5769 0.0003477 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
$model_114_CSS
z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1 -0.351702   0.294475 -1.1943   0.23235
ma1  0.237435   0.290139  0.8183   0.41316
ma2 -0.481718   0.077707 -6.1991 5.677e-10 ***
ma3 -0.212284   0.137227 -1.5470   0.12187
ma4  0.140748   0.084816  1.6594   0.09703 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
$model_114_ML
z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1 -0.388595   0.281648 -1.3797   0.16767
ma1  0.275024   0.278413  0.9878   0.32324
ma2 -0.479517   0.076916 -6.2343 4.539e-10 ***
ma3 -0.226626   0.131165 -1.7278   0.08402 .
ma4  0.135026   0.086423  1.5624   0.11820
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
$`model_114_CSS-ML`
z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1 -0.388706   0.281550 -1.3806   0.16740
ma1  0.275135   0.278312  0.9886   0.32287
ma2 -0.479540   0.076914 -6.2348 4.524e-10 ***
ma3 -0.226674   0.131130 -1.7286   0.08388 .
ma4  0.135023   0.086431  1.5622   0.11824
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
$model_310_CSS
z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1 -0.114459   0.076995 -1.4866   0.1371
ar2 -0.422290   0.070851 -5.9602 2.519e-09 ***
ar3 -0.089112   0.077166 -1.1548   0.2482
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
$model_310_ML
z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1 -0.112964   0.077047 -1.4662   0.1426
ar2 -0.418351   0.070420 -5.9408 2.837e-09 ***
ar3 -0.087789   0.076714 -1.1444   0.2525
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
$`model_310_CSS-ML`
z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1 -0.112977   0.077048 -1.4663   0.1426
ar2 -0.418335   0.070420 -5.9406 2.84e-09 ***
ar3 -0.087801   0.076714 -1.1445   0.2524
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
$model_211_CSS
z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1  0.147460   0.192159  0.7674   0.4429
ar2 -0.399352   0.077451 -5.1562 2.52e-07 ***
ma1 -0.273855   0.214609 -1.2761   0.2019
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$model_211_ML
z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1  0.148792   0.194411  0.7653   0.4441
ar2 -0.395596   0.077386 -5.1120 3.188e-07 ***
ma1 -0.273818   0.217564 -1.2586   0.2082
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

$`model_211_CSS-ML`
z test of coefficients:

      Estimate Std. Error z value Pr(>|z|)
ar1  0.148180   0.194455  0.7620   0.4460
ar2 -0.395649   0.077342 -5.1156 3.128e-07 ***
ma1 -0.273026   0.217648 -1.2544   0.2097
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```